DEEP LEARNING-BASED PEDIATRIC PNEUMONIA CLASSIFICATION FROM X-RAY IMAGES

Jason Chiu, Jason Tieh, Raghavi Ramasamy, Waqas Ur Rehman

Schulich School of Engineering, University of Calgary

ABSTRACT

Pneumonia remains a leading cause of mortality in children under five worldwide [1], and early, accurate detection, particularly via chest X-ray, is critical for improving outcomes, yet distinguishing it from other respiratory conditions is challenging due to overlapping visual features and image variability. Machine learning-based detection systems have the potential to assist clinicians by improving diagnostic consistency and enabling faster, more accessible screening, particularly in resource-limited settings. This study proposes a deep learning–based framework evaluating four state-of-the-art architectures—ResNet50, DenseNet121, ConvNeXtV2, and SwinV2—trained on a balanced dataset of 5,982 images, followed validation and testing on a cohort of 878, and 879 images respectively. By tuning hyperparameters and optimizer configurations [2], the study demonstrates the performance of the above models. Overall, the proposed approach provides a reliable, scalable solution to assist radiologists in clinical decision-making and supports the development of robust computer-aided diagnostic systems for respiratory diseases.

Index Terms— Pneumonia, Chest X-ray (CXR), Deep Learning, Convolutional Neural Networks (CNN), Swin Transformer V2, ConvNeXtV2, ResNet50, DenseNet121, Grad-CAM,Transfer Learning, ROC AUC, F1-score

1. INTRODUCTION

Developing accurate pneumonia detection systems is often hindered by the scarcity of expert-labeled data and the high cost of manual radiologist interpretation. Deep learning has emerged as a powerful tool for this task, utilizing Convolutional Neural Networks (CNNs) to capture local spatial features and Transformer-based architectures to model long-range contextual dependencies. However, while prior work demonstrates strong performance of individual architectures, fewer studies examine trade-offs between performance and interpretability, particularly in pediatric datasets where data scarcity and variability are more pronounced, and where high accuracy alone does not ensure clinically meaningful reasoning. These systems facilitate real-time classification, offering a second opinion to reduce diagnostic errors and improve clinical efficiency. This study evaluates the efficacy of ResNet50, DenseNet121, ConvNeXtV2, and SwinV2 architectures. By training these models on a balanced dataset of 5,982 images and performing a comparative analysis of optimized hyperparameters and performance metrics—including accuracy, F1-score, and ROC AUC—this study identifies the most robust frameworks for being adapted for automated pneumonia detection. In particular, this study highlights the trade-off between predictive performance and interpretability, emphasizing that higher accuracy does not necessarily correspond to clinically meaningful feature learning.

2. METHODS

2.1. Datasets

To train our models, a publicly available dataset of pediatric anterior–posterior (AP) chest X-ray images for pneumonia detection [3], accessed via Kaggle [4] was used. The dataset consists of labeled radiographs categorized into pneumonia and normal classes, and has been commonly adopted as a benchmark for evaluating deep learning models in medical image classification tasks.

To ensure robust model evaluation and generalization, the dataset was partitioned into training, validation, and test subsets using a 70–15–15 split. The validation and test sets were constructed to preserve the natural class imbalance present in the original dataset, providing a realistic estimate of model performance in clinical like conditions. Specifically, each of the validation and test sets contains 641 pneumonia images, along with 237 and 238 normal images, respectively.

The training set consists of 2991 pneumonia images and 1108 normal images. To mitigate class imbalance and improve model generalization, data augmentation techniques were applied exclusively to the training set. A total of 1883 augmented normal-class samples were generated to approximate class balance. The augmentation pipeline included random horizontal flipping (probability = 0.5), vertical flipping (probability = 0.1), random rotations within ±15°, affine transformations with translations up to 5% in both spatial dimensions and scaling in the range of 0.9–1.1, as well as photometric adjustments via brightness and contrast jitter (±20%). These transformations are commonly used in med-

ical imaging tasks to improve robustness to positional and acquisition variability [5].

A fixed augmented training dataset was generated prior to training. All models were trained on this dataset instead of applying dynamic augmentation, eliminating variability and ensuring performance differences are due to architectural characteristics rather than data exposure.

2.2. Model Architectures

The four models represent different generations of deep learning design. DenseNet121 [6] and ResNet50 [7] are CNN-based architectures using dense and residual connections, relying on batch normalization and depth to capture global context. ConvNeXt V2 [8] modernizes CNNs with transformer-inspired features such as large-kernel depthwise convolutions and layer normalization, while preserving local inductive bias. Swin Transformer V2 [8] departs further by modeling image patches as tokens and applying self-attention within shifted windows to capture long-range dependencies. All models were pretrained on ImageNet and adapted for binary classification with a sigmoid output head.

As summarized in Table 1, key differences lie in normalization and scale. ResNet50 and DenseNet121 use batch normalization, which can be unstable with small batch sizes, whereas ConvNeXt V2 and Swin V2 use layer normalization, offering more stable fine-tuning. In terms of size, DenseNet121 (8M) and ResNet50 (25M) are smaller than ConvNeXt V2 and Swin V2 (88–89M), increasing capacity but also the risk of overfitting on small pediatric datasets.

2.3. Experimental setup and procedure

Each model was developed by a different group member, who selected hyperparameters and training strategies with the objective of maximizing individual model performance. This approach reflects realistic model development workflows, where architectural performance is influenced by both design and optimization choices; however, it introduces variability in hyperparameter selection that limits strict isolation of architectural effects. All models were initialized using ImageNet-pretrained weights to leverage transfer learning, which has been shown to significantly improve performance in medical imaging tasks with limited data. Final classification layers were modified to output predictions for binary classification (pneumonia vs. normal).

Input image sizes were set to 224×224 for DenseNet121, ConvNeXtV2, and ResNet50, and 256×256 for Swin V2, with standard normalization applied based on ImageNet statistics. Batch sizes ranged from 16 to 32, and training was conducted for 10–30 epochs depending on the model. Optimization strategies varied: DenseNet121 and ResNet50 used Adam, while ConvNeXtV2 and Swin V2 used AdamW. Learning rates ranged from 1×10^{-5} to 3×10^{-4}, with weight decay applied in select models. Learning rate scheduling via ReduceLROnPlateau and early stopping (patience = 5) were applied where appropriate to improve convergence and mitigate overfitting.

A staged training strategy was used for ConvNeXtV2, where the backbone was unfrozen after 5 epochs to enable fine-tuning. ResNet50 additionally incorporated dropout (0.5) for regularization. All models were trained using either CrossEntropyLoss or BCEWithLogitsLoss depending on output formulation.

To ensure a controlled comparison, all models were trained and evaluated on the same fixed dataset split and augmented training set. Performance was assessed using standard classification metrics, including accuracy, precision, recall, and F1-score, on validation and test data. Consequently, while all models were trained on identical data splits, differences in optimization strategies may act as confounding factors when interpreting comparative performance. All results are based on single training runs per model, and performance variance across runs was not evaluated.

3. RESULTS

The following details provides a comprehensive analysis of experimental results obtained from evaluating four state-of-the-art deep learning architectures: ResNet50, DenseNet121, ConvNeXtV2, and SwinV2. Re-running the models may result in minor fluctuations in the final performance metrics. However, given the consistent convergence observed across all architectures and the use of standardized hyperparameter settings, any such variations are expected to be negligible and do not alter the comparative rankings or the overall conclu-

Table 1. Comparison of Transfer Learning Architectures for Pediatric Pneumonia Classification

Property	**ResNet50**	**DenseNet121**	**ConvNeXt V2**	**Swin V2**
Architecture type	CNN (residual)	CNN (dense)	Modernized CNN	Vision Transformer
Connectivity	Skip connections	Dense block reuse	Depthwise 7×7 conv	Shifted window attention
Normalization	Batch Norm	Batch Norm	Layer Norm	Layer Norm
Parameters	$\sim$25M	$\sim$8M	$\sim$89M	$\sim$88M
Global context	Implicit (depth)	Implicit (depth)	Near-global	Explicit (attention)
Input size	224×224	224×224	224×224	256×256

Table 2. Performance comparison of models

Model	Accuracy	Precision	Recall	F1-score	ROC AUC
SwinV2	0.9795	0.9796	0.9795	0.9794	0.9972
ConvNeXtV2	0.9783	0.9789	0.9783	0.9785	0.9971
DenseNet121	0.9772	0.9772	0.9772	0.9772	0.9933
ResNet50	0.9704	0.9720	0.9704	0.9707	0.9931

sions of this study. The diagnostic reliability of the models was assessed on a dedicated test set consisting of 879 images. A variety of metrics, including Accuracy, Precision, Recall, F1-score, and ROC AUC, were employed to provide a comprehensive view of model performance [9].

The experimental results indicate that SwinV2 achieved the highest overall performance across almost all evaluation categories, likely due to its ability to model long-range spatial dependencies through self-attention mechanisms. SwinV2 recorded a test accuracy of 0.9795 and an F1-score of 0.9794. ConvNeXtV2 followed closely with an accuracy of 0.9783, suggesting that modernized CNN designs can approach transformer-level performance while retaining convolutional inductive biases, while DenseNet121 and ResNet50 achieved 0.9772 and 0.9704, respectively. The ROC AUC (Receiver Operating Characteristic Area Under the Curve) provides a threshold-independent measure of the model's discriminative ability. SwinV2 achieved an ROC AUC of 0.9972, marginally outperforming ConvNeXtV2 at 0.9971. The traditional CNN architectures, DenseNet121 and ResNet50, achieved ROC AUC scores of 0.9933 and 0.9931, respectively.

In clinical practice, the cost of a false negative (failing to diagnose pneumonia) is typically higher than the cost of a false positive (unnecessary follow-up for a healthy patient), as the former can lead to rapid disease progression and mortality [10]. Therefore, high recall for the pneumonia class is a critical requirement for any computer-aided diagnostic system.

SwinV2 demonstrated the highest sensitivity for pneumonia, achieving a recall of 0.9938 for the pneumonia class. This means that the model correctly identified over 99.3 percent of actual pneumonia cases in the test set. ConvNeXtV2 followed with a pneumonia recall of 0.9782, while DenseNet121 achieved 0.9844. ResNet50, despite having the lowest overall accuracy, maintained a pneumonia precision of 0.9936, but its recall was significantly lower at 0.9657, indicating a higher likelihood of missing actual cases compared to the other models.

While qualitative image analysis is typically required to fully assess Grad-CAM, the experimental layout assigned specific layers for attention visualization. For SwinV2, the window-based self-attention mechanism produces hierarchical feature maps that emphasize global contextual patterns, which is essential for identifying the diffuse infiltrates characteristic of viral pneumonia. Comparative studies have noted that while DenseNet121 consistently focuses on core pathological regions, ResNet50 sometimes exhibits "scattered" attention, highlighting peripheral areas or anatomical structures such as the ribs that are not directly related to pneumonia. The use of modern architectures like ConvNeXtV2, with its large 7×7 receptive fields, facilitates a more holistic view of the thoracic cavity, aligning the model's "attention" more closely with the diagnostic logic used by human radiologists.

Table 3. Class-wise performance metrics

Model	Class	Prec.	Rec.	F1	Sup.
SwinV2	NORMAL	0.9825	0.9412	0.9614	238
	PNEUMONIA	0.9785	0.9938	0.9861	641
ConvNeXtV2	NORMAL	0.9433	0.9789	0.9608	238
	PNEUMONIA	0.9921	0.9782	0.9850	641
DenseNet121	NORMAL	0.9578	0.9578	0.9578	238
	PNEUMONIA	0.9844	0.9844	0.9844	641
ResNet50	NORMAL	0.9141	0.9832	0.9474	238
	PNEUMONIA	0.9936	0.9657	0.9794	641

Table 4. Confusion Matrix for All Models

DenseNet121

	Predicted	
Actual	N	P
N	227	10
P	10	630

SwinV2

	Predicted	
Actual	N	P
N	224	14
P	4	637

ConvNeXtV2

	Predicted	
Actual	N	P
N	233	5
P	14	627

ResNet50

	Predicted	
Actual	N	P
N	234	4
P	22	619

The integration of these heatmaps [11] allows clinicians to verify that the model is identifying legitimate radiographic markers of disease—such as consolidations, air bronchograms, or pleural effusions—rather than relying on image artifacts or medical devices (e.g., pacemakers or tubes) often found in complex clinical cases.

4. DISCUSSION

ResNet50: Although achieving high accuracy, Grad-CAM results show attention to peripheral regions rather than lung fields, suggesting reliance on non-relevant features. **DenseNet121:** Demonstrates both high accuracy and clinically meaningful attention, with consistent focus on central lung regions indicative of effective feature learning. **ConvNeXtV2:** Exhibits sparse and scattered activations with limited focus on lung structures, which may contribute to weaker performance, especially for normal cases. **SwinV2:** Produces diffuse and inconsistent attention patterns; despite strong quantitative performance, its lack of localized focus raises concerns about interpretability and clinical reliability. [12]

Overall, the Grad-CAM analysis highlights that quantitative performance alone is insufficient, as models with similar

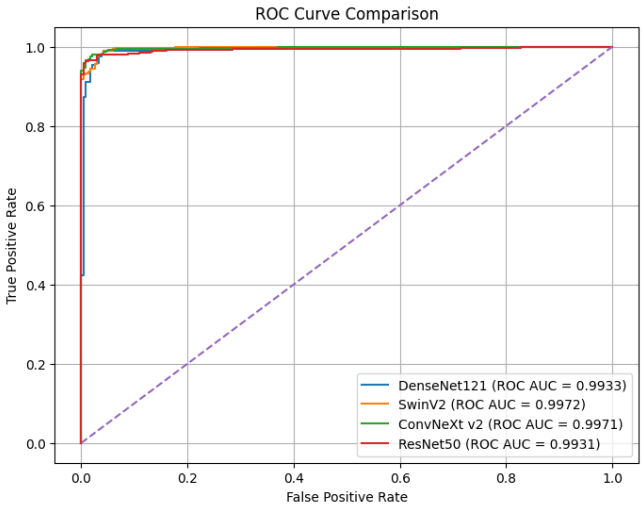

Fig. 1. Comparison of ROC Curves for the best developed models of the four architectures.

accuracy can differ significantly in clinical validity and feature relevance. While several models achieved high classification accuracy, their attention patterns varied significantly in clinical relevance. DenseNet121 demonstrated the most reliable and interpretable behavior, consistently focusing on anatomically meaningful lung regions, suggesting it may be more suitable for clinical deployment despite marginally lower performance. In contrast, ResNet50, SwinV2, and ConvNeXtV2 exhibited varying degrees of attention to non-

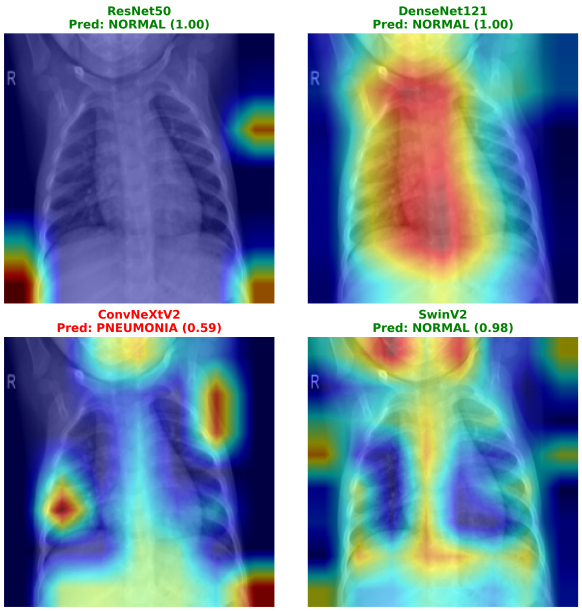

Fig. 2. Sample Grad-CAM visualization for a classified normal chest X-ray case.

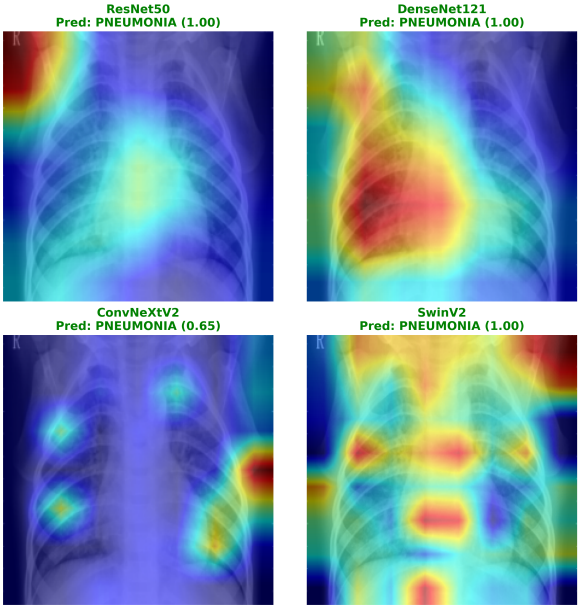

Fig. 3. Sample Grad-CAM visualization for pneumonia classified chest X-ray case.

relevant or diffuse regions, raising concerns about potential reliance on spurious features. These findings emphasize the need to evaluate models using both performance metrics and interpretability to ensure clinically meaningful decision-making. A key limitation of this study is the lack of standardized hyperparameter tuning across models, which introduces confounding factors in attributing performance differences solely to architecture. Additionally, reliance on a single pediatric dataset without external validation limits generalizability and may increase susceptibility to dataset-specific biases.

5. CONCLUSION

This study evaluated transfer learning across four deep learning architectures — ResNet50 [7], DenseNet121 [6], ConvNeXt V2 [8], and Swin Transformer V2 [13] — for binary classification of pediatric pneumonia from chest X-rays, finding that ImageNet-pretrained models effectively bridge the domain gap to medical radiographs. While lightweight CNNs like DenseNet121 remain competitive despite fewer parameters, transformer-based architectures offer greater representational capacity at the cost of more careful regularization. Grad-CAM [14] visualizations confirmed that model activations localized to clinically relevant lung regions, and future work could explore ensemble methods, bacterial-vs-viral multi-class classification, and evaluation on larger, more diverse pediatric cohorts.

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